Chatty-KG Appendix

Classify Question (I_C)

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Classify a given question as either "self-contained"  or "non-self-contained" while considering
the context of resolving pronouns and references. In this task, "self-contained" questions are
those that can be understood and answered without needing additional context or information, even
when they refer to entities or concepts mentioned in prior conversation. "Non-self-contained"
questions are those that, due to the presence of pronouns or references to prior conversation,
require additional context or information to be answered correctly.

Question: {question}
Classification:
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Rephrase Ambiguous Question(I_{RF})

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Given the following conversation and a follow up question, rephrase the follow up question to be
a standalone question. Preserve the original question in the answer sentiment during rephrasing.

Chat History: {chat_history}
Follow Up Input: {question}
Standalone question:
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Question Understanding (I_{QU})

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Extract triples from the given question by following these steps carefully.
**Let’s think step by step.**

### **Step 1: Extract Key Entities**
- Identify explicit entities mentioned in the question.
- If an entity is missing but implied, introduce a variable (e.g., ?var).

### **Step 2: Identify the Predicates**
- **Prioritize nouns that represent the core relationship being questioned.** These nouns often
describe the key property or attribute being asked about (e.g., "capital," "author," "moons").
- **Explicitly exclude generic verbs like "has," "is," "are," "was," "were," "do," "does,",
"did" when a more specific noun or verb can express the relationship.**
- If no such core noun exists, then consider other verbs, excluding the helper verbs listed above.
- Extract the most relevant predicate from the question.
- The predicate should be specific and relates two entities.

### **Step 3: Construct the Triples**
- Formulate the triple using the extracted subject, predicate, and object.
- If the object is unknown, use a variable (e.g., ?answer).
- Ensure at least one meaningful triple is produced.

### **Step 4: Output the Result**
- Return the final output as a JSON object in this exact format: ‘{{ "triples": [] }}‘.
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**Let’s apply this step-by-step approach to extract the correct triple.**

### **Examples:**
#### **Example 1**
**Question:** Who wrote Pride and Prejudice?
**Step 1 - Extract Entities:** ["Pride and Prejudice", "?author"]
**Step 2 - Identify Predicates:** "wrote"
**Step 3 - Construct Triples:** ‘[ ["?author", "write", "Pride and Prejudice"] ]‘

#### **Example 2**
**Question:** What is the capital of France?
**Step 1 - Extract Entities:** ["France", "?capital"]
**Step 2 - Identify Predicates:** "capital"
**Step 3 - Construct Triples:** ‘[ ["France", "capital", "?capital"] ]‘

### **Input:**
Question: {question}

### **Response:**
‘‘‘json
```

Vertex Selection (I_{VL})

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#### Instruction:
Given an entity and a list of labels, return the most similar label.
1. If an exact match (case-insensitive) exists in the list, return that label immediately and
exclusively, ignoring any other labels.
2. If no exact match is found, select the most semantically similar label based on meaning,
context, and common associations.

**Important Notes**:
- An **exact match** means the entity must match a label character-for-character, ignoring case.
- Do not modify, append, 3or alter the exact-match label.
- If an exact match exists, it must be returned even if a semantically similar label is also
present.
- Prioritize labels that closely match the entity meaning, not just syntactic similarity.

Only Return the label in a **JSON object** in the following format: {{"value": label}}.
Do not add any explanations.

### Inputs:
Entity: {entity}
List of labels: [{vertex_label_list}]

### Response:‘‘‘json
```

Query Selection (I_{QS})

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### Instruction:
Let’s think step by step about what the question is asking and what the answer would look like.
- Evaluate each keyword relevance to retrieving the correct answer. Focus on keywords that are
directly related to answering the question.
- Only include keywords that directly contribute to retrieving the correct answer.
- Exclude keywords related to the item or context, and focus solely on those that provide the key
information needed for the answer.
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Only choose keywords from the provided list - do not generate new ones.
Return the keywords in a JSON object in the following format:
{{ "keywords": [] }}.

### Inputs:
Question: {question}
List of Keywords: [{predicate_list}]

### Response:```json
```

Final Answer Reformalization I_{FA}

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Given a question and its structured answer extracted from the Knowledge Graph. Convert the
structured answers to an equivalent natural language answer suitable for human conversation

Question: {question}
Answers: {answers}

Reformalized Answer:
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